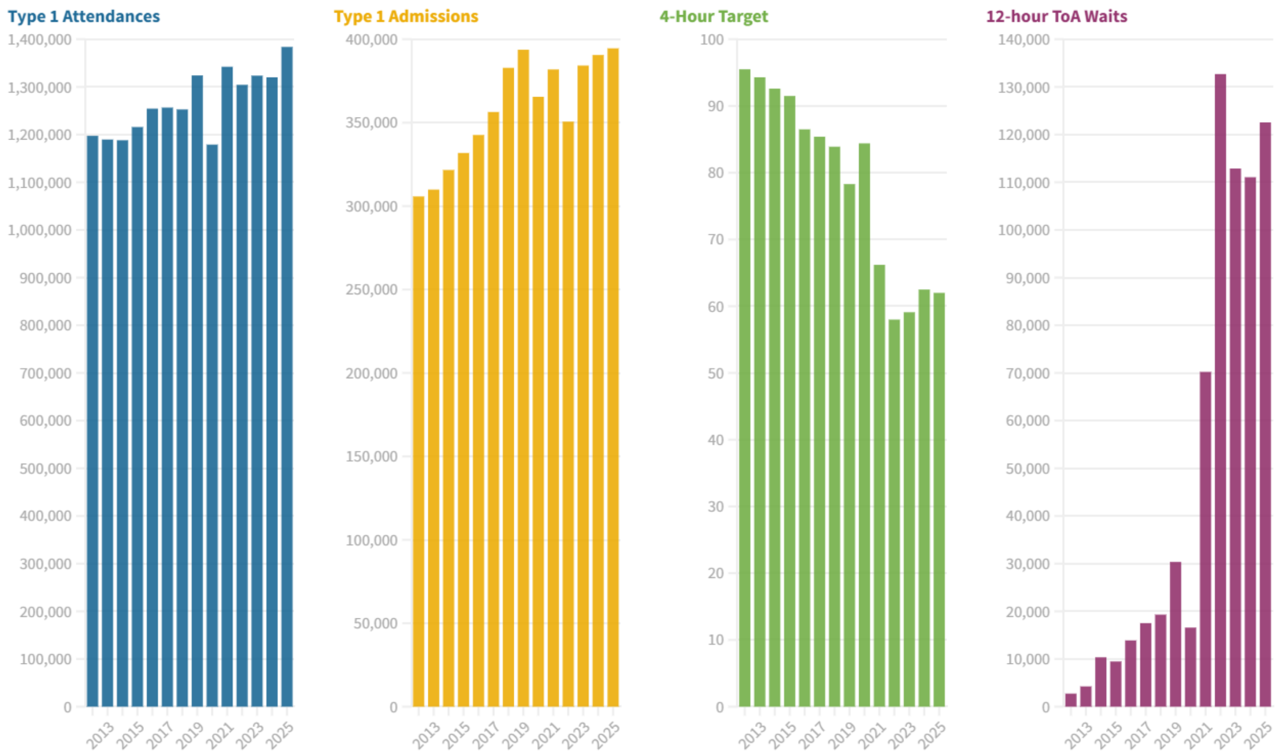
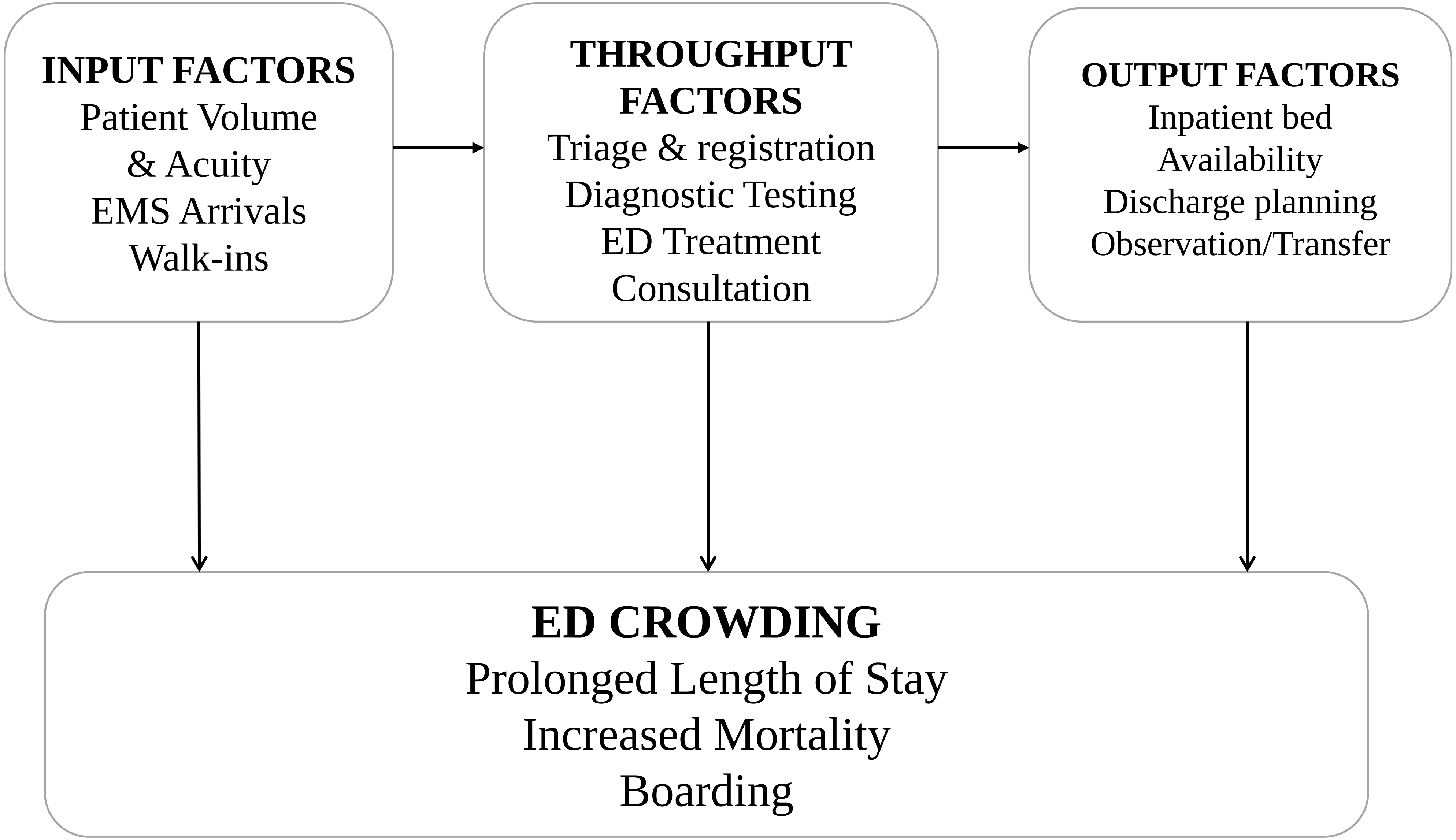
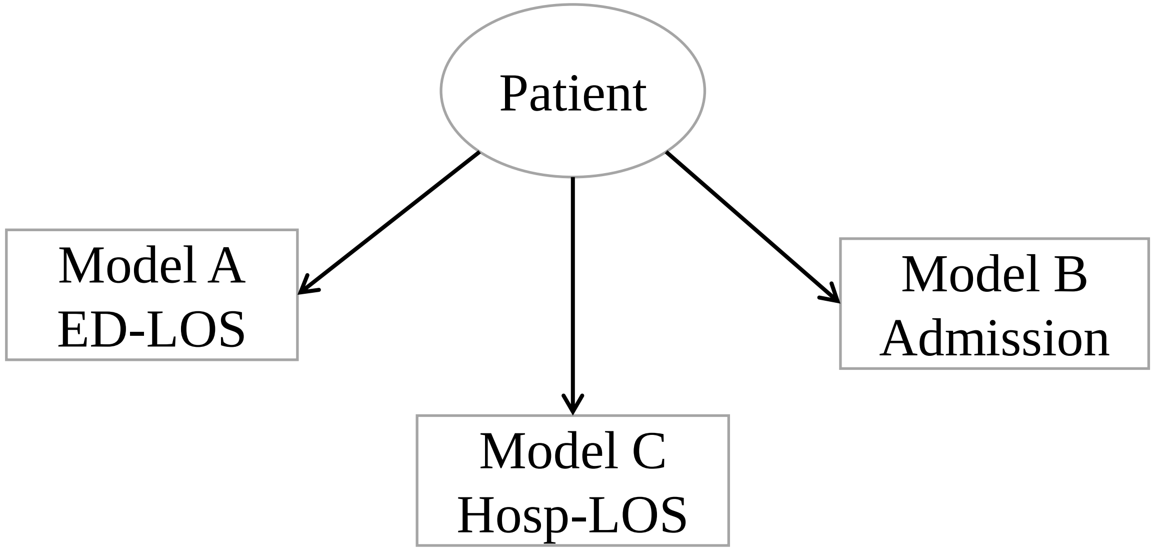
August Performance Figures since 2012

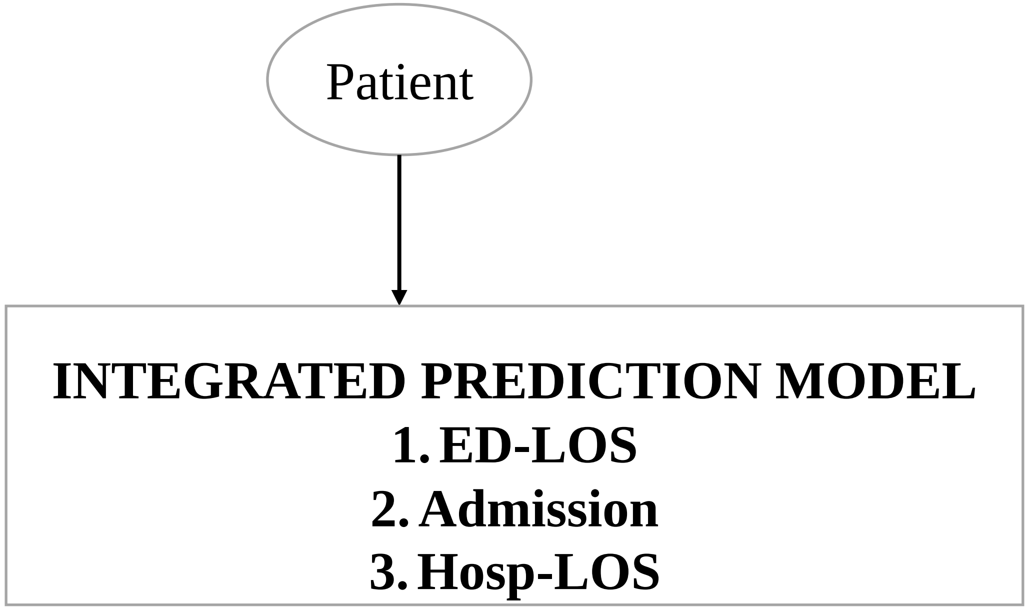



INPUT FACTORS
Patient Volume
& Acuity
EMS Arrivals
Walk-ins
THROUGHPUT
FACTORS
Triage & registration
Diagnostic Testing
ED Treatment
Consultation
OUTPUT FACTORS
Inpatient bed
Availability
Discharge planning
Observation/Transfer
ED CROWDING
Prolonged Length of Stay
Increased Mortality
Boarding

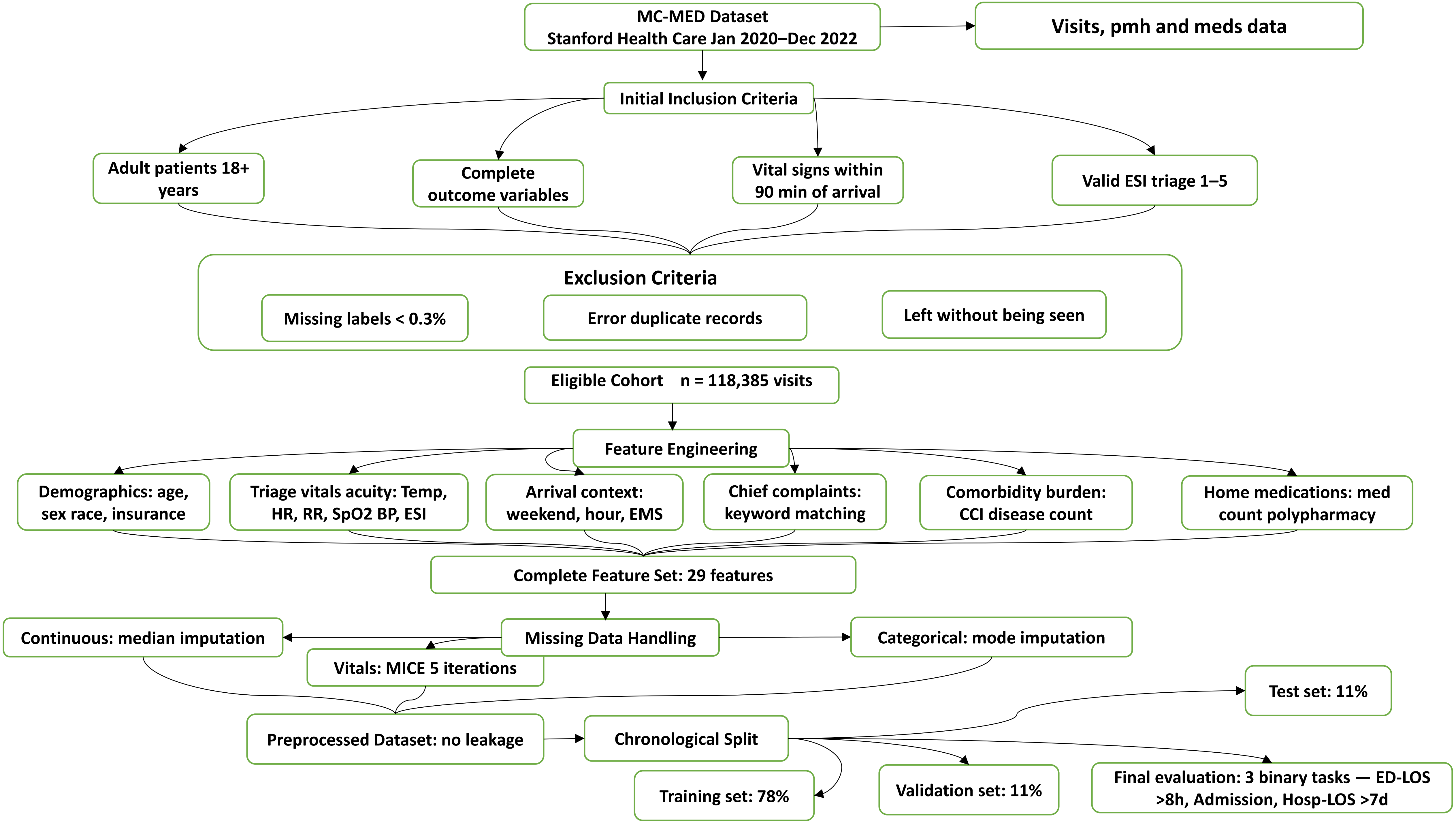
Fragmented vs. Complete Interrelated Trajectory

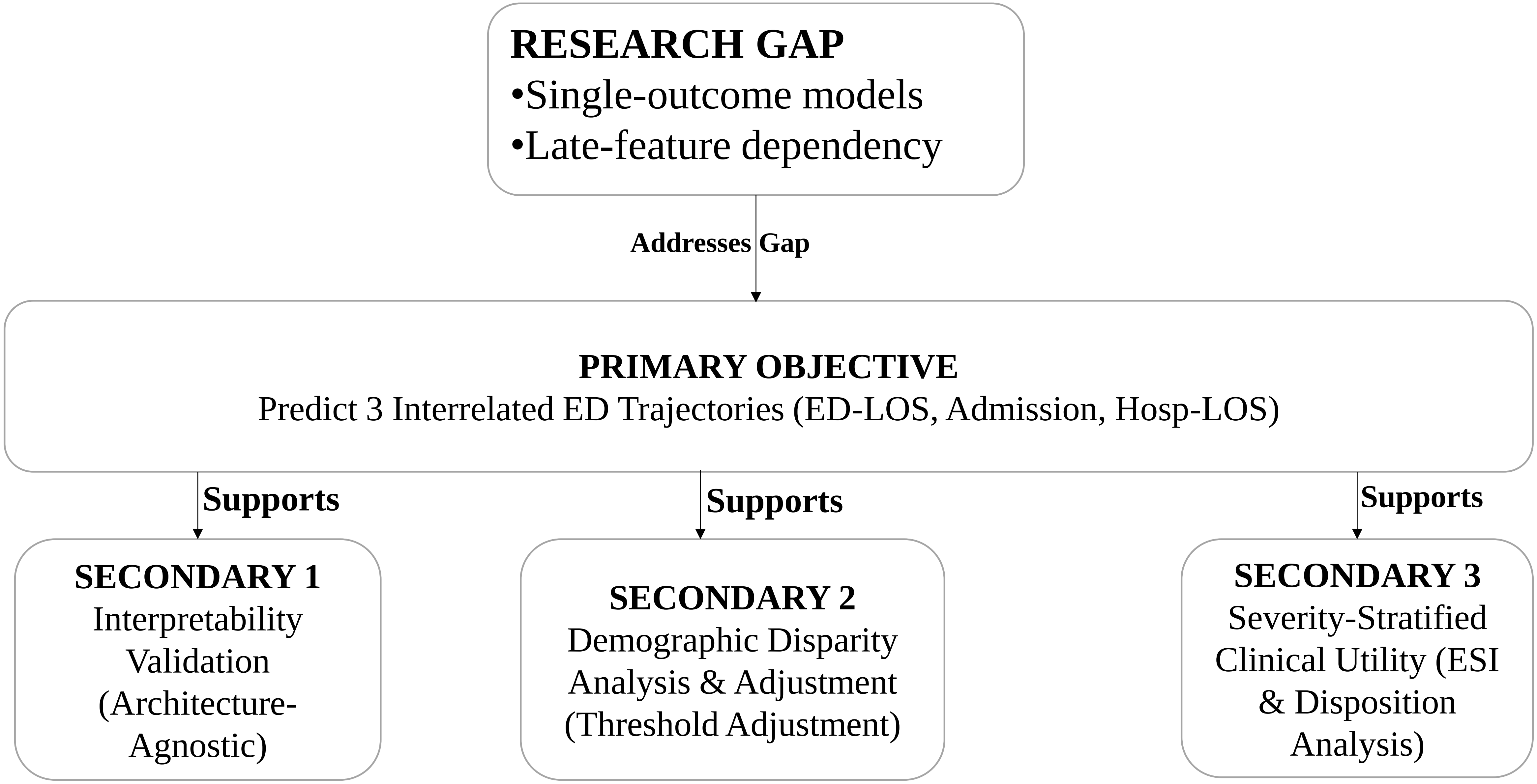


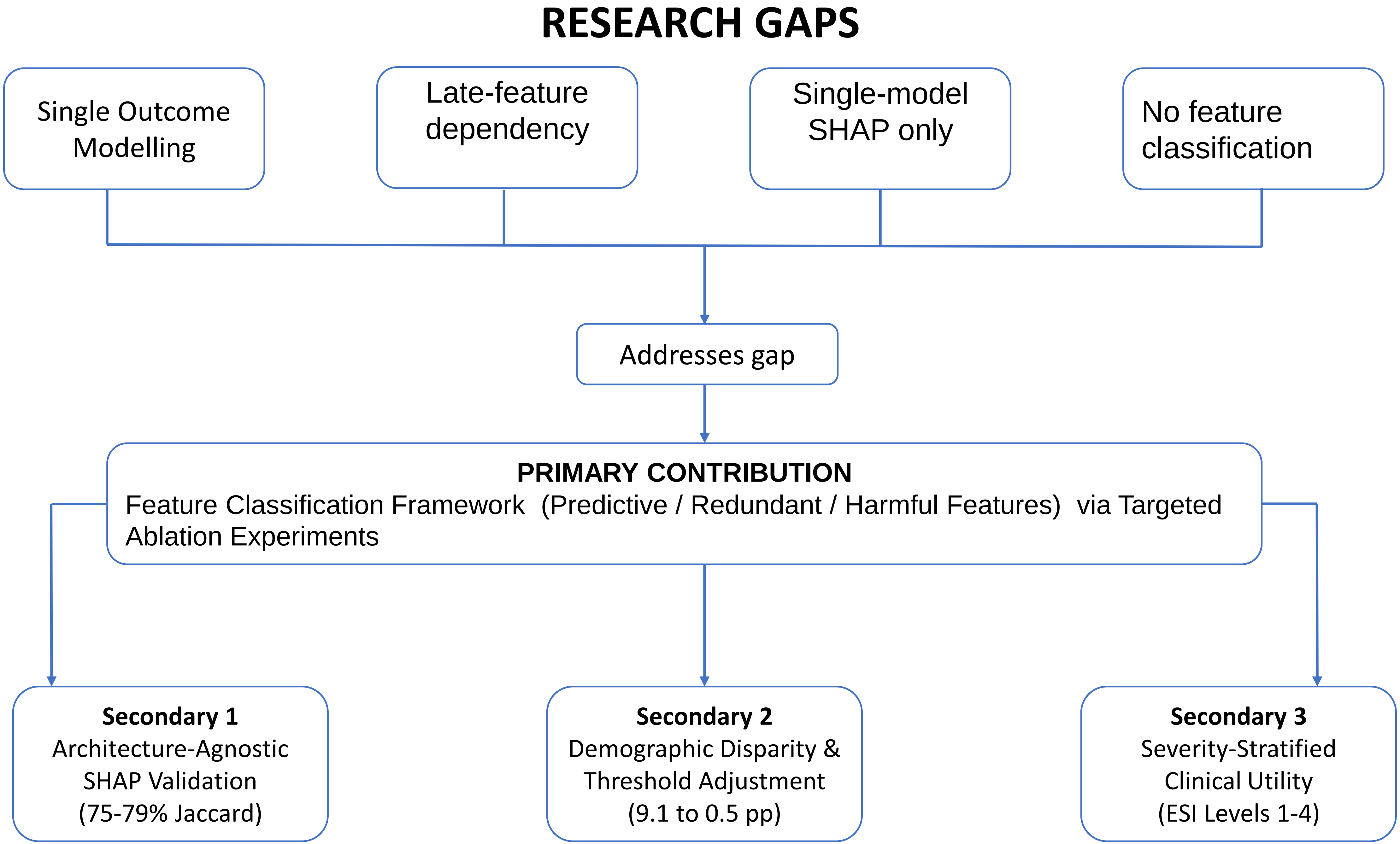
Single Patient Trajectory is Fragmented



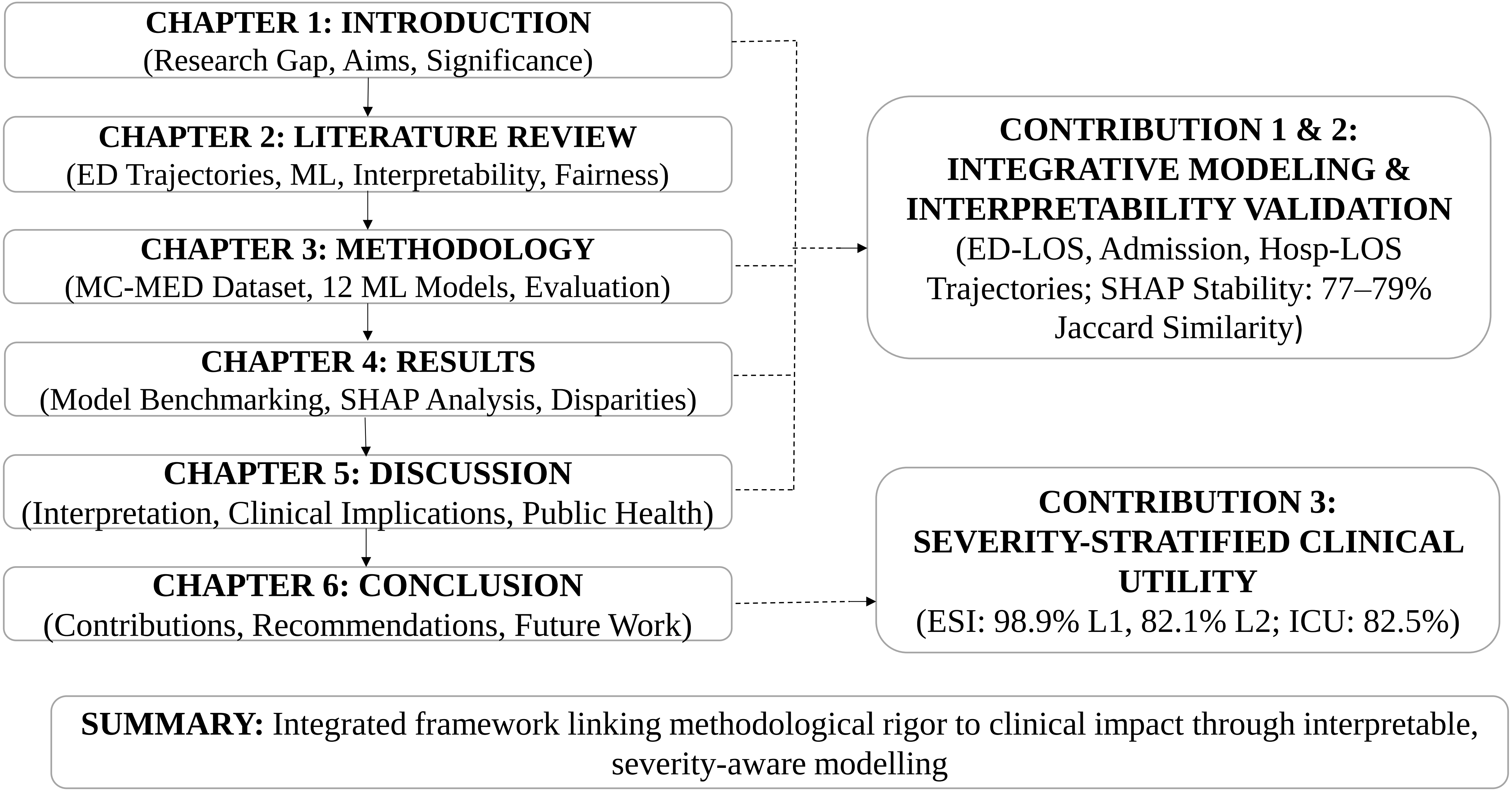
Complete Interrelated Trajectory


MC-MED Dataset
Stanford Health Care Jan 2020–Dec 2022
Visits, pmh and meds data
Initial Inclusion Criteria
Adult patients 18+ years
Complete outcome variables
Vital signs within 90 min of arrival
Valid ESI triage 1–5
Exclusion Criteria
Missing labels < 0.3%
Error duplicate records
Left without being seen
Eligible Cohort n = 118,385 visits
Feature Engineering
Demographics: age, sex race, insurance
Triage vitals acuity: Temp, HR, RR, SpO2 BP, ESI
Arrival context: weekend, hour, EMS
Chief complaints: keyword matching
Comorbidity burden: CCI disease count
Home medications: med count polypharmacy
Complete Feature Set: 29 features
Missing Data Handling
Continuous: median imputation
Vitals: MICE 5 iterations
Categorical: mode imputation
Preprocessed Dataset: no leakage
Chronological Split
Test set: 11%
Training set: 78%
Validation set: 11%
Final evaluation: 3 binary tasks — ED-LOS >8h, Admission, Hosp-LOS >7d


RESEARCH GAP
•Single-outcome models
•Late-feature dependency
Addresses Gap
PRIMARY OBJECTIVE
Predict 3 Interrelated ED Trajectories (ED-LOS, Admission, Hosp-LOS)
Supports
Supports
Supports
SECONDARY 1
Interpretability Validation (Architecture-Agnostic)
SECONDARY 2
Demographic Disparity Analysis & Adjustment (Threshold Adjustment)
SECONDARY 3
Severity-Stratified Clinical Utility (ESI & Disposition Analysis)


RESEARCH GAPS
Single Outcome Modelling
Late-feature dependency
Single-model SHAP only
No feature classification
Addresses gap
PRIMARY CONTRIBUTION
Feature Classification Framework (Predictive / Redundant / Harmful Features) via Targeted Ablation Experiments
Secondary 1
Architecture-Agnostic SHAP Validation
(75-79% Jaccard)
Secondary 2
Demographic Disparity & Threshold Adjustment
(9.1 to 0.5 pp)
Secondary 3
Severity-Stratified Clinical Utility
(ESI Levels 1-4)

STRUCTURE AND KEY CONTRIBUTIONS



CONCEPTUAL FRAMEWORK: INTEGRATIVE PREDICTIVE MODELING of Interrelated Clinical Trajectories in Emergency Care